Author: Micheal Lanham

SECTION 1

Fundamentals of LLM Reasoning & Planning

- **Decomposition & Planning:** Reasoning requires two distinct cognitive operations: decomposition (breaking a problem down into smaller subproblems) and planning (determining the sequence, dependencies, and tool execution strategy for those subproblems).
- **Reasoning Scaffolding:** While non-reasoning models require explicit prompting techniques to think step-by-step, frontier reasoning models handle step-by-step deliberation natively.
- **Chain-of-Thought (CoT):** Directs an LLM to generate explicit, step-by-step intermediate thoughts before outputting a final answer, which improves performance on complex logic tasks.
- **The ReAct Paradigm (Reason, Act, Observe):** Alternates between internal natural-language reasoning steps and external tool executions in an interactive loop. ReAct uses observations from tool outputs to update reasoning dynamically and adjust subsequent actions.
- **Strategic Planning:** Moves beyond single-question handling by generating explicit, multi-step plans stored externally in memory or context, allowing the agent to track progress across long-horizon goals.

SECTION 2

Instructing Agents to Reason and Plan

- **CoT Agent Implementation:** Embeds step-by-step reasoning instructions into the agent's system prompt to force explicit intermediate calculations.
- **ReAct Agent Implementation:** Equips the agent with specialized tool functions (such as time-jump functions) and instructs it to reason, execute a tool call, inspect the observation, and reflect before proceeding.

SECTION 3

Advanced Reasoning Strategies

- **Tree-of-Thought (ToT):** Explores multiple reasoning paths in parallel, evaluates candidate solution steps, prunes unpromising branches, and backtracks when a path fails. It is effective for complex search or strategy tasks but incurs higher token and latency costs.
- **Reflexion:** A self-critique pattern where a solver agent's output is evaluated by a critic agent. The critic generates feedback hints that are injected back into the solver's context for iterative retries without updating model weights.
- **Pattern Selection:** Matches specific strategies to task requirements—CoT for logic puzzles, ReAct for tool-heavy retrieval, ToT for deep strategic search, and Reflexion for iterative self-correction.

SECTION 4

Sequential Thinking MCP Server

- **External Reasoning Scratchpad:** Uses Anthropic's reference Model Context Protocol (MCP) server, which provides a dedicated `SequentialThinking` tool.
- **State Continuity:** The server does not perform reasoning itself; instead, it provides persistent external storage for thought steps that the agent rereads at each step to maintain continuity across turns.
- **Combining Reasoning Patterns:** Enables agents to integrate CoT planning, ReAct tool execution, ToT branching, and Reflexion feedback into a unified reasoning architecture.

SECTION 5

Key Design Best Practices

- **Tool Scoping:** Tool lists should be kept lean (fewer than 10 tools per agent) to prevent context inflation and tool selection errors during ReAct loops.
- **Guardrails & Limits:** Hard turn caps and typed input/output validation are required to prevent infinite execution loops and validate intermediate plan outputs.